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UNIVERSITY OF RWANDA
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WEB DESIGN TECHNICAL REPORT 2025-2026

Hospital Appointment System

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1. Introduction

1.1 Historical Background

Hospitals traditionally relied on paper-based appointment booking. This caused long queues and scheduling conflicts. A web-based Hospital Appointment System digitizes appointment management.

1.2 Problem Statement

Manual appointment booking wastes time, causes double bookings, and makes patient records difficult to manage.

1.3 Objectives

- Develop an online appointment system.
- Allow patient registration/login.
- Allow doctors to manage appointments.
- Enable appointment booking and cancellation.
- Improve efficiency.

1.4 Expected Outcomes

Easy online booking,

reduced waiting time,
secure records,
improved communication.

1.5 Project Rationale

Improves healthcare service delivery and reduces administrative workload.

2. System Analysis and Design

2.1 Intended Users

Patients, Doctors, Administrators.

2.1.2 System Partners

Hospitals, Health staff. patients, receptionists

2.2 High-Level Architecture

patient -> web browser -> hospital appointment system - database

-doctor module

-admin modil

-patient module

2.2.2 UI Design

Pages: Home, Patient Registration, Login, Doctor Registration, Doctor Login, Appointment Booking, Dashboard, Contact.

2.2.3 Database Design

Entities: Patients, Doctors, Appointments, department, doctorschedule, medical record.

Relationships:

Patient books Appointment.

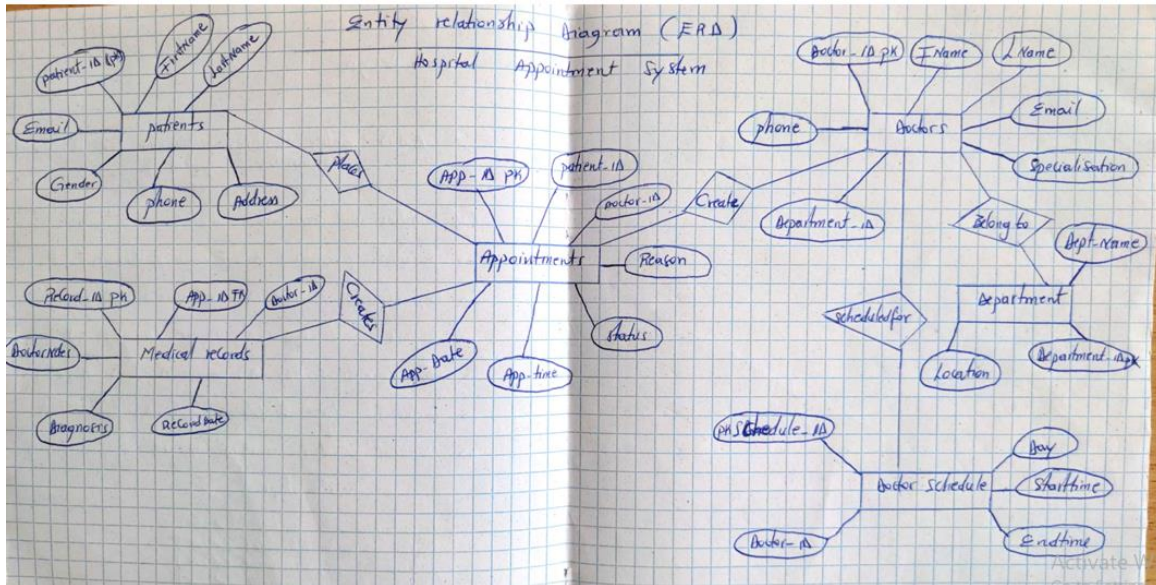
Doctor manages appointment

Doctor belong in department.

Appointment make medical record.

Doctor has doctor schedule

ERD diagram



3. Implementation

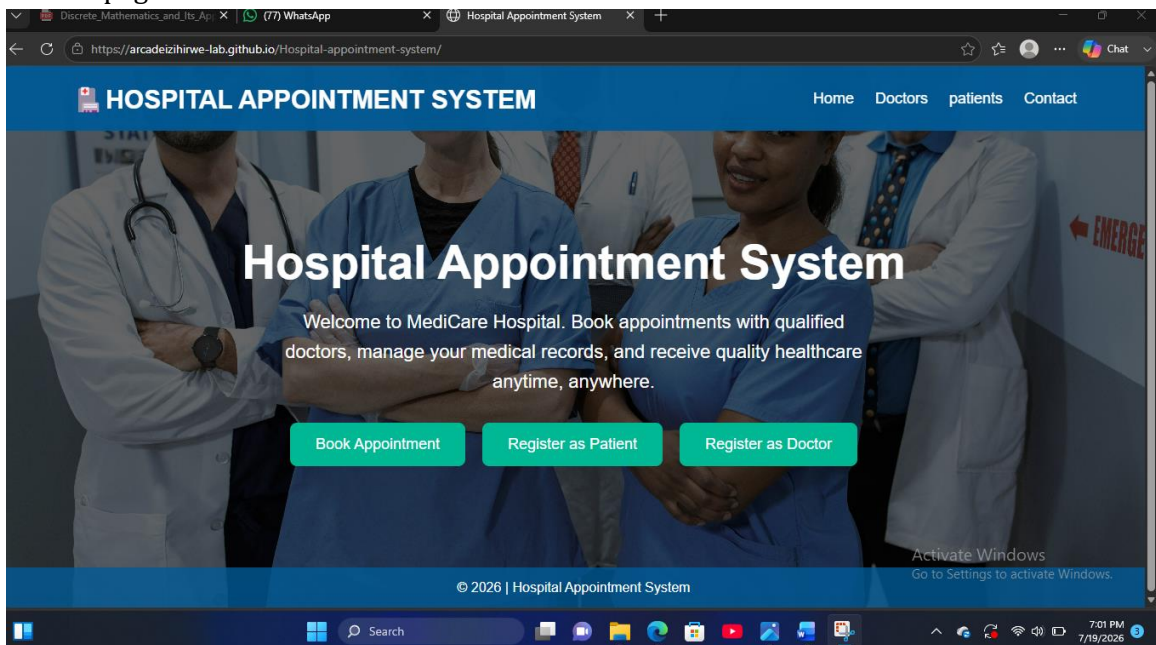
3.1 Introduction

The system was developed using HTML, CSS, JavaScript, PHP and MySQL.

3.2 Screenshots

Include screenshots of:

- Home page



This is the front door of the system. Its main job is to get you to the right place quickly.

- Patient Registration

Welcome to quickcare appointment system

Your Health, Our Priority

Welcome!

We are pleased to have you at quickcare appointment system. Our dedicated team of professionals is here to provide you with the best healthcare services in a comfortable and caring environment.

To help us serve you better, please click the button below to make Appointment

[Book Appointment](#)

Please select the medical service you need. This will help us recommend the hospitals and doctors that can best treat your condition.

please select the healthcare you need .

[General Consultation](#)

This is usually the first form a new patient fills out so the hospital can safely create their official Medical Record.

- Doctor Registration

HOSPITAL APPOINTMENT SYSTEM

Home Doctors patients Contact

Doctor Registration

Create your professional account

First Name Last Name

Email Phone

Gender Specialization

Medical License Number

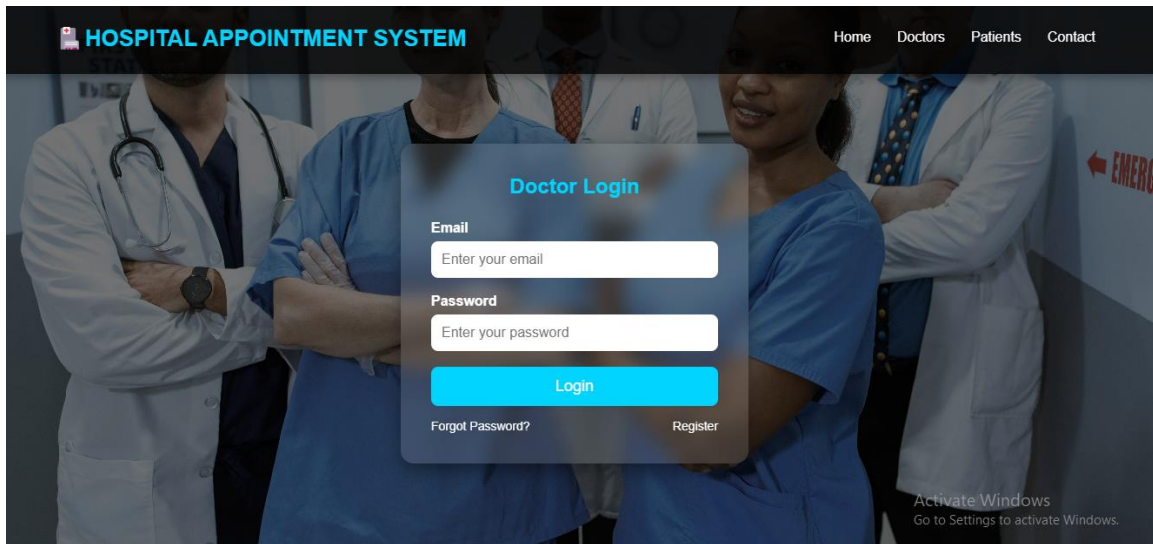
Hospital Address

Join Our Medical Team

Register as a certified doctor to manage patient appointments, view schedules, and provide better healthcare services through our Hospital Appointment System.

Unlike a patient registration page, a **Doctor Registration** page focuses on verification, qualifications, and scheduling. It is usually restricted or reviewed by hospital administrators to ensure that only legally certified medical professionals can access the system.

- doctor sign



Help doctor to sign in hi /her account after creating account to see his /her appointments. This is where the doctor proves who they are to the system.

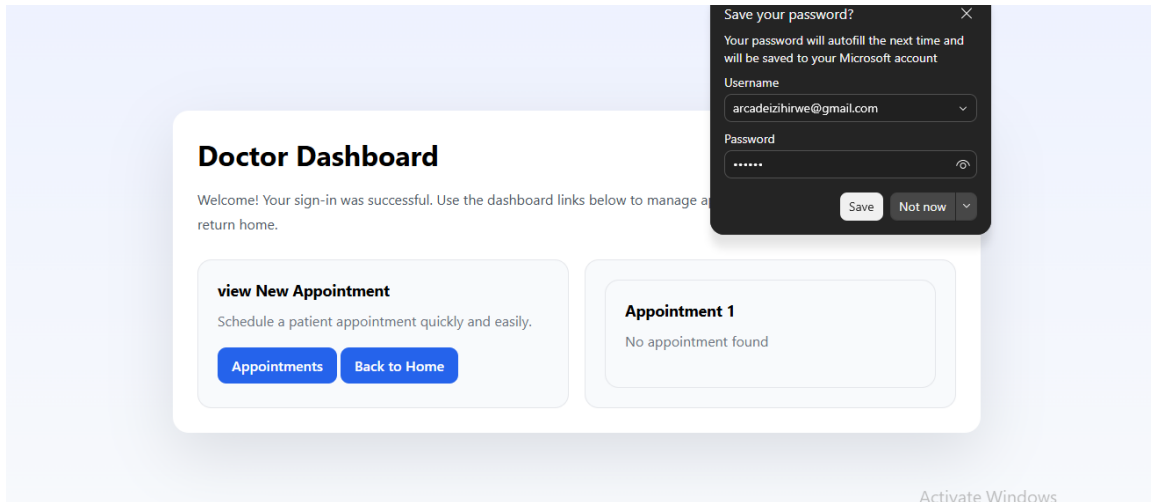
- **Username / Email:** The doctor enters their hospital-issued email address or a unique staff ID number.
- **Password:** A strong, alphanumeric password created during their registration.

- Appointment Booking

The image shows a web application titled "quickcare appointment system" with the tagline "Book Your Appointment Easily". The main content area features an "Appointment Form" with the following fields: "Full Name" (placeholder: "Enter your full name"), "Email Address" (placeholder: "Enter your email"), "Phone Number" (placeholder: "Enter your phone number"), "Select Department" (a dropdown menu with "Choose Department" selected), and "Select Doctor" (a dropdown menu with "Choose Doctor" selected). The background is a blurred image of medical professionals. An "Activate Windows" watermark is visible in the bottom right corner.

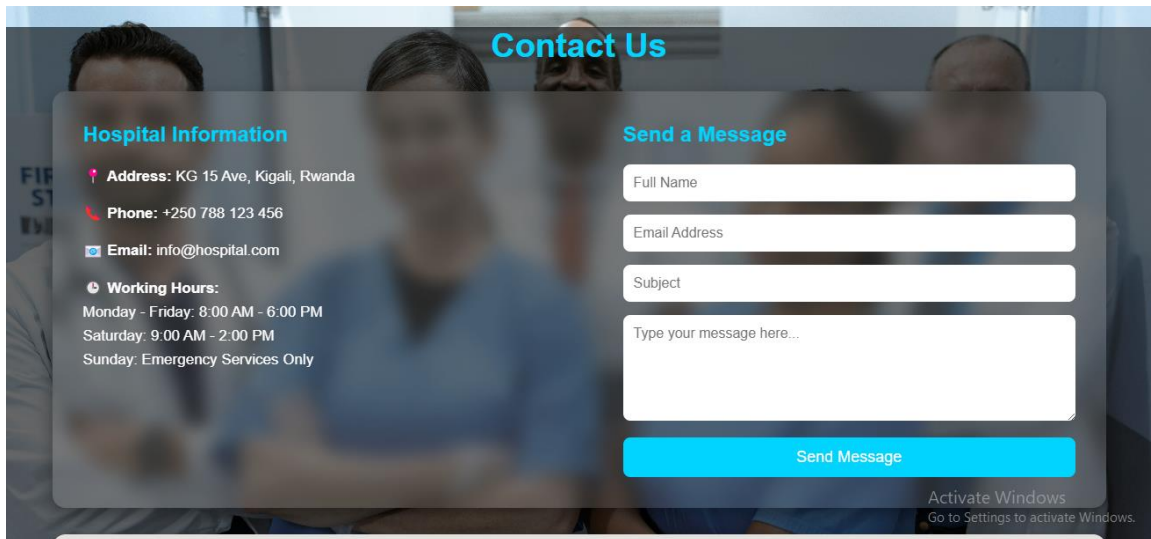
Help patient to book appointments. The **Book Appointment Page** is the heart of the entire patient portal. This form turns a complicated scheduling process into a few simple clicks.

- Doctor Dashboard



This is where doctor manages his /her appointments. It is designed to give them a complete, quick look at their entire workday so they can focus on treating patients rather than clicking through menus.

- Contact Page



The **Contact Page** serves as the safety net of the hospital appointment system.

4. Conclusion and Recommendation

The Hospital Appointment System achieved its main objectives by providing online appointment scheduling and management.

Recommendations:

- Add SMS/Email notifications.
- Integrate online payment.
- Train users.
- Deploy on a secure server.

Appendix

GitHub Repository Link: <https://arcadeizihirwe-lab.github.io/Hospital-appointment-system/>

Username: arcadeizihirwe-lab